The major benefits of an AMM are that it is always available and that a traditional counterparty is not necessary to execute a trade. These provisions are very important for smart contracts and DeFi development because of the guarantee that a user can exchange assets at any moment if necessary. Users maintain custody of their funds until they complete the trade; hence, counterparty risk is zero. An additional benefit is *composable liquidity*, which means any exchange contract can plug into the liquidity and exchange rates of any other exchange contract. AMMs make this particularly easy because of their guaranteed availability and their allowance of one-sided trading against the contract. Composable liquidity correlates comfortably with the concept of DeFi Legos (which we will discuss later).

One drawback to an AMM is *impermanent loss*: the opportunity–cost dynamic between offering assets for exchange and holding the underlying assets to potentially profit from the price movement. The loss is impermanent because it can be recovered if the price reverts to its original level. To illustrate, consider two assets, A and B, each initially worth 1 ETH as in Figure 4.5. The AMM contract holds identical quantities of 100 of each asset and naively offers both at a fixed exchange rate of 1:1. We use ETH as the unit of account to track the contract's return on its holdings and any impermanent loss. At the given balances and market exchange rates, the contract has 200 ETH in escrow. Suppose asset B's price appreciates to 4 ETH in the wider market and asset A's price appreciates to 2 ETH. Arbitrageurs exchange all of asset B